

46-937: Final Project Report Rubric

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1 Report (due Wednesday Oct 7th by 11:59 PM ET)

You will submit reports as a group assignment on Gradescope, which asks you to list all the group members when you submit; this will make it easy for us to update the gradebook. For the Gradescope submission, you need to submit a PDF file that consists of your main report as well as a code appendix containing all of your relevant code.

You should not have any code visible in your report. The report should have the following structure, which differs slightly from a typical IMRaD format, since **each member of the group will have to write a section about their approach for submitting predictions individually**:

- Introduction (Group Grade)
- Group Member #1 Predictions Overview (Individual Grade)
- Group Member #2 Predictions Overview (Individual Grade)
- Group Member #3 Predictions Overview (Individual Grade)
- Group Member #4 Predictions Overview (Individual Grade - if 4 members)
- Final Predictions Overview (Group Grade)

- Conclusion (Group Grade)

The group portion of the report is 100 points, and each individual's Group Member Prediction Overview sections are also worth 100 points.

The following sections details the grading rubrics for each part of the report, where **Basic items** are all or nothing while **Detailed items** are graded on a scale.

1.1 Introduction (Group Grade: 20 points)

Describes the prediction problem you're tackling based on the provided data dictionary (on Canvas) and context [from the original Kaggle page](#) (the counts are wrong since I made fake data from this). Also gives a brief summary of your results.

- **Basic item (5pts):** Does the report include an *Introduction*?
- **Basic item (5pts):** Does the *Introduction* section describe the problem you're tackling.
- **Basic item (5pts):** Does the *Introduction* section give a brief summary of the results.
- **Detailed item (rated 0 to 5):** What is the quality of this summary? Do I know what type of model you ultimately settled on as a group when submitting your final predictions?

1.2 Group Member Predictions Overview (Individual Grade: 100 points)

Each member of the group will need to write a section of the report that details how they came up with their individual predictions that were submitted on Friday September 25th. This individual section should include the following:

1.2.1 Data Prep (15 points)

Describes any pre-processing of the dataset that was considered, such as transformations or how you handled missing values:

- **Basic item (5pts):** Does your individual prediction overview include a *Data Prep* subsection?
- **Basic item (5pts):** Does the *Data Prep* section include a description of any pre-processing steps?
- **Detailed item (rated 0 to 5):** What is the quality of the description? Are you preprocessing steps appropriate?

1.2.2 Method Selection (85 points)

Describes the models/methods you considered when coming up with your approach for submitting predictions on Friday Sept 25th. You will need to compare three different approaches:

- Naive baseline model - just using the average of each response variable as the prediction, ignoring all other predictors
- Not deep learning model - any type of model or models that are NOT a form of deep learning that we considered in this class
- Deep learning model - some type of deep learning model(s) covered in this class!

You will need to demonstrate why you ultimately chose whatever approach you considered based on an appropriate model selection procedure (e.g., some form of cross-validation).

- **Basic item (5pts):** Does your individual prediction overview include a *Method Selection* subsection?
- **Basic item (10pts):** Does your *Method Selection* subsection contain a description of some type of model(s) that is NOT deep learning?
- **Detailed item (rated 0 to 10):** What is the quality of this description? Is this an appropriate model to consider in comparison to the naive baseline and some type of deep learning approach?
- **Basic item (10pts):** Does your *Method Selection* subsection contain a description of some type of deep learning approach?
- **Detailed item (rated 0 to 10):** What is the quality of this description? Did you provide justification for your considered architecture(s)?
- **Basic item (10pts):** Does the *Method Selection* subsection contain a description for how you will compare/evaluate the different approaches for generating predictions relative to a naive baseline?
- **Detailed item (rated 0 to 10):** What is the quality of this description? How appropriate is your comparison/evaluation approach for this particular prediction task? How did you handle tuning the hyperparameters (if any) of the considered approaches?
- **Basic item (10pts):** Does the *Method Selection* subsection contain the results from running your considered procedure for comparing the different approaches to generate predictions?
- **Detailed item (rated 0 to 10):** How are these results presented? Did you clearly state which approach performed best with some notion of uncertainty accounted for?

Again, **each member will need to write their own Group Member Prediction Overview section!**

1.3 Final Predictions Overview (Group Grade: 65 points)

Describes the approach your group settled on for submitting predictions on Friday Oct 2nd. You will need to provide some type of objective comparison between the approach you settled on as a group with each of your considered individual approaches, and also comment on whether or not the individual predictions leaderboard impacted your decision. **There should only be one section like this in your report since it is shared across your group.** Additionally, **your final approach for generating predictions may or may not be one of your individual member's approaches.** It's reasonable if you decide as a group to come up with an entirely new approach, BUT you will need to compare this new approach with each of your individual approaches in some way.

- **Basic item (5pts):** Does your report include a *Final Predictions Overview* section?
- **Basic item (10pts):** Does this section contain a description for how you will compare/evaluate the different considered approaches that were used for generating predictions? This includes each of the individual member approaches from the earlier deadline, along with any new approaches that the group came up with.
- **Detailed item (rated 0 to 10):** What is the quality of this description? How appropriate is your comparison/evaluation approach for this particular prediction task? How did you handle tuning the hyperparameters (if any) of the considered approaches?
- **Basic item (10pts):** Does this section contain the results from running your considered procedure for comparing the different approaches to generate predictions?
- **Detailed item (rated 0 to 10):** How are these results presented? Did you clearly state which approach performed best with some notion of uncertainty accounted for?
- **Basic item (10pts):** Does this section discuss whether or not the individual predictions leaderboard impacted their selection?
- **Detailed item (rated 0 to 10):** What is the quality of this discussion? Is your assessment appropriate and justified?

1.4 Discussion (15 points)

Gives your conclusions and summarizes what you have learned with regards ultimately generating these predictions. Are there any limitations with the approaches you used? What do you think are the next steps to follow-up your project?

- **Basic item (5pts):** Does the report include a *Discussion* section?
- **Basic item (5pts):** Does the report discuss limitations of the analyses in the report?

- **Basic item (5pts):** Does the report discuss future work based on their results and limitation that could be completed in a follow-up?